Preliminaries

Review

- The expression
$$\frac{f(a+h)-f(a)}{h}$$
gives the _____ rate of change of $f(x)$ between $x =$ _____ and $x =$ _____.
- The *derivative* of a function at a point $x = a$ is
- $\lim_{x\to c} f(x) = L$ means we get _____ as close to _____ as we want by taking x _____. This is the value we *expect* $f(x)$ to take at $x = c$, even if it doesn't.
- Therefore, $f'(a)$ is what we *expect* the rate of change of the function to be at $x = a$.

The Derivative Function

When we evaluated the derivative of $f(x) = x^2$ at $a = 2$, we carried out the following calculation:

$$\begin{aligned} f'(2) &= \lim_{h\to 0} \frac{(2+h)^2 - 2^2}{h} \\ &= \lim_{h\to 0} \frac{4+4h+h^2-4}{h} \\ &= \lim_{h\to 0} \frac{4h+h^2}{h} \\ &= \lim_{h\to 0} 4+h = 4 \end{aligned}$$

To evaluate the derivative of the same function at $a = 4$, say, we carry out an almost identical calculation. This is pretty pointless. Instead, for a given function $f(x)$, we calculate the *derivative function*, $f'(x)$:

> **Definition:** Given a function $f(x)$, its *derivative function* is
> $$f'(x) = \lim_{h\to 0} \frac{f(x+h)-f(x)}{h}.$$

Note: The following are equivalent ways to denote the derivative of $y = f(x)$:

$$f'(x), \quad y', \quad \frac{dy}{dx}, \quad \frac{d}{dx}f(x)$$

Fill in the blank: If $f(x)$ is a function, then $f'(a)$ gives the _____ of $f(x)$ at the point $x = a$.

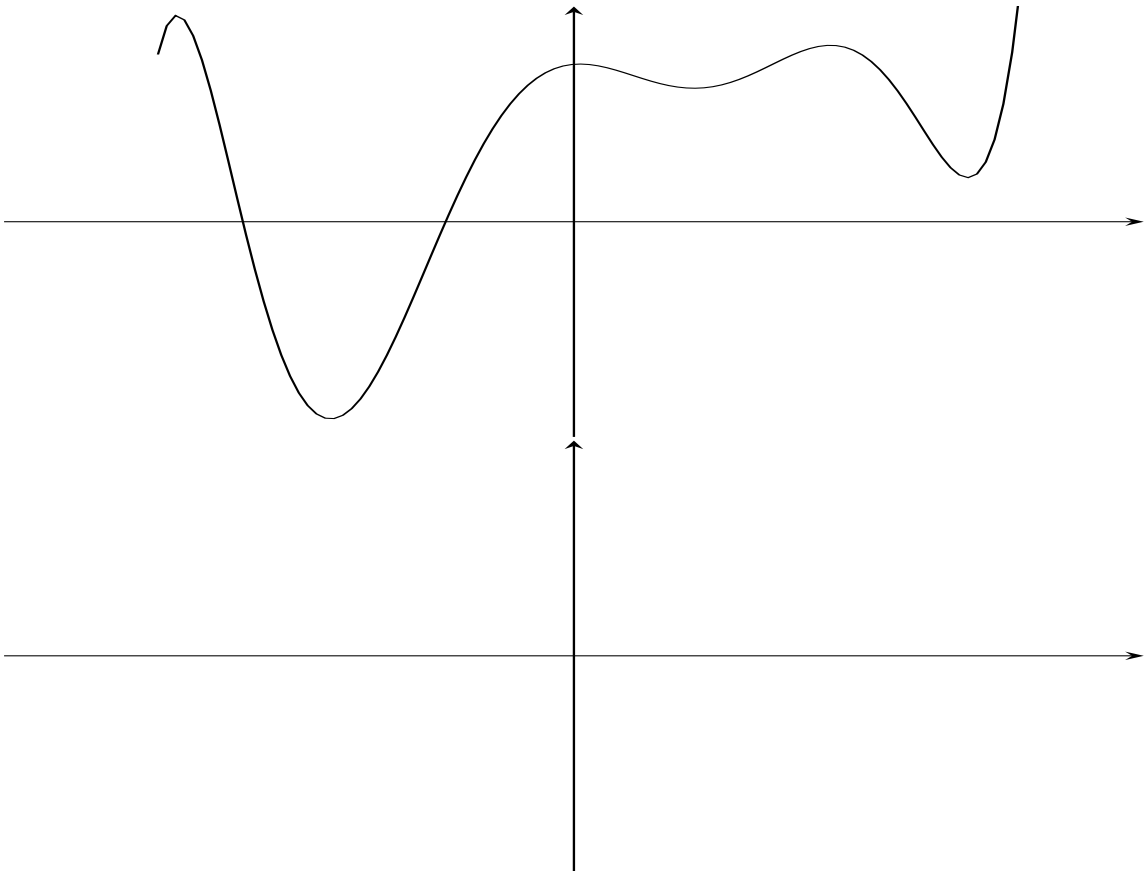
Graphing the Derivative Function

Since $f'(a)$ is the slope of $f(x)$ at $x = a$:

- When $f(x)$ is increasing, its slope (and therefore its derivative $f'(x)$) is _____.
- When $f(x)$ is decreasing, its slope (and therefore its derivative $f'(x)$) is _____.
- At a point $x = a$ where the graph of the $f(x)$ turns from increasing to decreasing (or vice versa), its slope must therefore be _____. Therefore at such a point, $f'(a) =$ _____.

Given a function $f(x)$, we can sketch its derivative $f'(x)$ using these ideas!

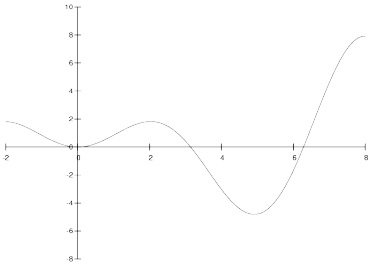
1. Examine the graph of a function $f(x)$ below. Sketch a graph of $f'(x)$ on the axes beneath. □Regrade



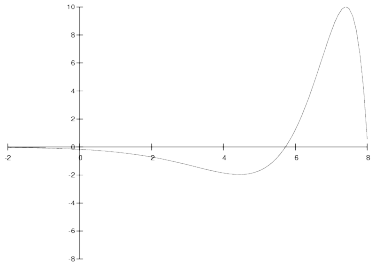
2. Match the following graphs on the left to their derivatives on the right.

□Regrade

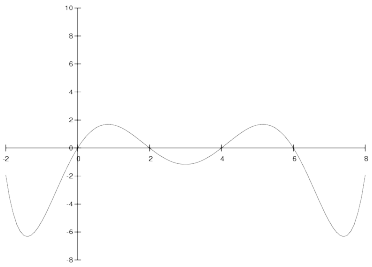
(a)



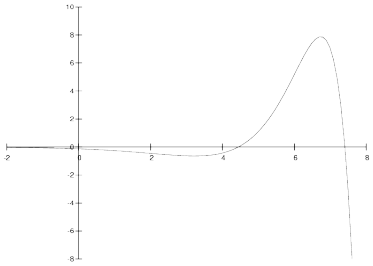
(b)



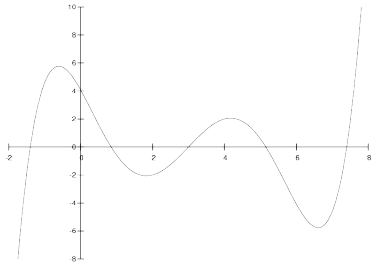
(c)



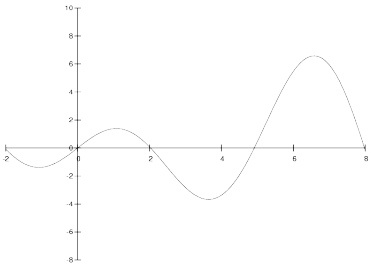
(1)



(2)



(3)



Interpreting the Derivative - Units

3. Suppose the function $s(t)$ gives the distance due East from Durham of a car traveling along the road.

□Regrade

(a) What are sensible units for t and for s?

(b) What are the units of the derivative function of $s(t)$, namely $s'(t)$?

(c) What would $s'(t)$ being negative mean about the motion of the car?

(d) Describe in words what knowing a formula for $s'(t)$ would tell you about the car. Include units.

Values of $f(x)$ and $f'(x)$

The *value of a function* $f(x)$ at a point $x = a$: what the function *is* at $x = a$.

The *value of the derivative* $f'(x)$ at $x = a$: what the function is *doing* at $x = a$.

4. Suppose that $P(t)$ gives the population of the United States in millions of people, □Regrade where t is 'years since 1800'.

(a) What does the expression $P(213) = 300$ mean? Include units.

(b) What does the expression $P'(213) = 2.1$ mean? Include units.

5. Suppose that $q(m)$ thousands of units of a particular product are sold when the price □Regrade is m dollars.

(a) Explain the following statements in words, giving units for all your numbers:

i. $q(10) = 100$.

ii. $q'(10) = -3$.

(b) Use the following statements to fill in the blanks:

i. If the price of the product is low (less than \$5 per item), increasing it increases demand, as people think it is of higher quality: If $m < \$5$, $q'(m)$_____.

ii. However, if the price is too high (greater than \$5 per item), increasing it further decreases demand, as people think it is too expensive: If $m > \$5$, $q'(m)$_____.

(c) If the data in part (b) of this question is true, what is $q'(5)$? Explain.

(d) If you were given a graph of $q'(m)$, explain how you would find the price that maximizes the number of units sold. Justify your answer.